

BIS Knowledge Base: Chemical Department (CHD)

Technical Standards, Sectional Committees, Certification Requirements & RAG Ingestion Guide

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Department Code: CHD (Chemicals)

Scope: National Standardization & Quality Control

System Ingestion Notice: This document is optimized for vector search embedding and RAG retrieval systems. Indian Standard (IS) numbers, exact titles, and functional categories are explicitly repeated across sections to preserve semantic context during text chunking.

1. Department Scope & Core Responsibilities

The **Chemical Department (CHD)** is one of the premier technical departments of the **Bureau of Indian Standards (BIS)**, operating under the oversight of the Chemical Division Council (CDC/CHD C). The department is responsible for national standardization, quality evaluation, safety benchmarking, and environmental risk mitigation across a vast array of chemical manufacturing and processing sectors in India.

Primary Responsibilities of BIS CHD:

- Standard Formulation:** Drafting, revising, amending, and reaffirming Indian Standards (IS) for chemicals, raw materials, intermediates, finished products, and testing methods.
- Quality Control Orders (QCO) Support:** Technical appraisal and drafting support for Ministries (e.g., Ministry of Chemicals and Fertilizers, Ministry of Commerce and Industry) to enforce mandatory BIS certification (ISI Mark / Scheme I).
- Occupational Health & Chemical Safety:** Establishing guidelines for handling hazardous chemicals, industrial gases, personal protective equipment (PPE), and safety in chemical processing plants.
- Environmental & Water Quality Standards:** Defining standardized test procedures and limits for industrial water quality, wastewater discharge, ambient air quality, and stack emissions.
- Harmonization with International Standards:** Aligning Indian Standards with ISO and IEC standards as a Participating (P) or Observer (O) member to facilitate international trade.

2. Sectional & Technical Committees (CHD Committees)

Standardization within CHD is executed through specialized Sectional Committees comprising representatives from industry, research institutions, regulatory bodies, and government ministries.

Committee ID	Sectional Committee Name	Technical Scope & Focus Areas
CHD 01	Inorganic Chemicals	Raw acids, bases, industrial salts, heavy chemicals (Caustic Soda, Soda Ash, Sulphuric Acid, Chlorine, Alum).
CHD 05	Electroplating & Photographic Materials	Electroplating salts, anodes, metal finishing chemicals, photographic processing chemicals.
CHD 06	Industrial Gases	Compressed, liquefied, and dissolved gases (Oxygen, Nitrogen, Argon, Hydrogen, Acetylene, Carbon Dioxide).
CHD 07	Chemical Hazards	Classification of hazardous chemicals, MSDS, chemical labeling, storage, transport safety, and accident prevention.
CHD 08	Occupational Safety & Health	Industrial safety equipment, respirators, protective clothing, occupational health hazards in factories.
CHD 09	Ceramicware	Vitreous china sanitaryware, ceramic tiles, tableware, technical ceramics, porcelain.
CHD 10	Glass, Glassware & Laboratoryware	Flat glass, container glass, laboratory glassware, volumetric glassware, thermal glass.

Committee ID	Sectional Committee Name	Technical Scope & Focus Areas
CHD 13	Water Quality for Industrial Purposes	Boiler feed water, cooling tower water, industrial effluent parameters, water treatment chemicals.
CHD 14	Printing Inks, Stationery & Allied	Offset inks, flexographic inks, writing inks, ball point pens, pencils, paper adhesives.
CHD 15	Paper & Paper Products	Writing & printing paper, packaging paper, pulp, tissue paper, testing methods for paper.
CHD 16	Paper Based Packaging Materials	Corrugated fibreboard boxes, paper bags, multiwall paper sacks, liquid packaging cartons.
CHD 17	Leather, Tanning Materials & Allied	Raw hides, tanned leather, tanning synthetic agents, footwear leathers, leather physical testing.
CHD 19	Footwear	Safety footwear, industrial boots, sports footwear, rubber/polymeric footwear, components.
CHD 20	Paints, Varnishes & Related Products	Architectural coatings, industrial enamels, primers, varnishes, powder coatings, thinners.
CHD 21	Raw Materials for Paints & Varnishes	Resins, pigments (Titanium Dioxide), extenders, solvents, driers, plasticizers.
CHD 23	Brushware, Polishes, Lac Products	Industrial brushes, paint brushes, floor polishes, shoe polishes, natural lac and shellac.
CHD 25	Soaps, Detergents & Surfactants	Toilet soaps, laundry soaps, synthetic detergent powders, liquid detergents, LABSA, surfactants.
CHD 26	Explosives & Pyrotechnics	Commercial blasting explosives, detonators, safety fuses, fireworks, pyrotechnic mixtures.
CHD 27	Thermal Insulation	Mineral wool, glass wool, expanded polystyrene (EPS), polyurethane foam (PUR/PIR), calcium silicate.
CHD 30	Nuclear Energy for Peaceful Uses	Radiological protection equipment, nuclear chemical processing standards, isotopes.
CHD 35	Air Quality	Ambient air monitoring methods, stack emission sampling, fugitive emission standards, gases.
CHD 36	Water Quality	Drinking water standards (IS 10500), packaged drinking water (IS 14543), mineral water (IS 13428).

3. Major Products & Primary Indian Standards (IS Numbers & Exact Titles)

Below is a consolidated repository of key Indian Standards managed under the Chemical Department (CHD), essential for industrial compliance and mandatory QCO enforcement.

3.1 Industrial Chemicals & Acids (CHD 01 & CHD 06)

IS Number	Exact Standard Title	Application / Domain
IS 252 : 2013	Caustic Soda, Pure and Technical - Specification	Alkali manufacturing, soap, textile processing.
IS 269 : 2015	Ordinary Portland Cement, 33 Grade - Specification	Construction materials (Cross-dept ref).
IS 266 : 1993	Sulphuric Acid - Specification	Fertilizers, chemical synthesis, batteries.
IS 251 : 1998	Soda Ash, Technical - Specification	

IS Number	Exact Standard Title	Application / Domain
		Glass manufacturing, detergents, chemicals.
IS 309 : 2005	Compressed Oxygen Gas - Specification	Industrial cutting/welding, medical gas.
IS 1047 : 2019	Polyvinyl Chloride (PVC) Resins - Specification	Plastics, pipes, cable insulation raw material.
IS 15030 : 2001	Toluene Diisocyanate (TDI-80) - Specification	Polyurethane foam production, coatings.

3.2 Water, Environmental Quality & Consumer Safety (CHD 36 & CHD 25)

IS Number	Exact Standard Title	Mandatory / QCO Status
IS 10500 : 2012	Drinking Water - Specification	National baseline water quality standard.
IS 14543 : 2024	Packaged Drinking Water (Other than Packaged Natural Mineral Water) - Specification	Mandatory ISI Certification (Scheme I).
IS 13428 : 2005	Packaged Natural Mineral Water - Specification	Mandatory ISI Certification (Scheme I).
IS 2888 : 2004	Toilet Soap - Specification	Personal care hygiene products.
IS 4955 : 2001	Household Powder Detergents - Specification	Consumer cleaning agents.

3.3 Paints, Protective Coatings & Building Materials (CHD 20, CHD 09, CHD 10)

IS Number	Exact Standard Title	Application / Domain
IS 15489 : 2004	Plastic Emulsion Paints - Specification	Architectural wall finishes.
IS 2553 (Part 1) : 2018	Safety Glass - Specification: Part 1 General Purpose	Architectural, automotive, structural glazing.
IS 2553 (Part 2) : 2019	Safety Glass - Specification: Part 2 For Road Transport	Mandatory ISI Certification for automobiles.
IS 2556 (Part 2) : 2004	Vitreous Sanitary Appliances (Vitrenus China) - Part 2 Specific Requirements of Wash-Basins	Building construction & plumbing.
IS 13630 (Parts 1-15)	Ceramic Tiles - Methods of Test	Determination of water absorption, strength, wear.

4. Important Testing Standards & Analytical Methods

Testing standards established by CHD define rigorous physical and chemical test protocols required for third-party compliance verification in BIS laboratories.

- **IS 3025 (Multiple Parts) — Methods of Sampling and Test (Physical and Chemical) for Water and Wastewater:**
 - *IS 3025 (Part 11):* Determination of pH value using glass electrode meter at **25 °C**.
 - *IS 3025 (Part 16):* Determination of Total Dissolved Solids (TDS) gravimetrically after drying at **180 °C**.
 - *IS 3025 (Part 23):* Determination of Alkalinity by titration against standardized **H_2SO_4 / HCl**.
 - *IS 3025 (Part 44):* Determination of Heavy Metals (Lead, Cadmium, Arsenic) using Atomic Absorption Spectrometry (AAS) or ICP-MS.
- **IS 101 (Multiple Parts) — Methods of Sampling and Test for Paints, Varnishes and Related Products:**
 - *IS 101 (Part 1 / Sec 5):* Determination of volatile matter and non-volatile content at **105 °C ± 2 °C**.

- *IS 101 (Part 3 / Sec 2)*: Measurement of drying time (surface dry and hard dry).
- *IS 101 (Part 8 / Sec 5)*: Determination of lead content and heavy metals restrictions in decorative paints.
- **IS 516 (Chemical Testing aspects)**: Chemical analysis of cement and hardened concrete components (Chloride and Sulphate content).
- **IS 2088 : 1983 — Methods for Determination of Arsenic**: Silver diethyldithiocarbamate spectrophotometric method.

5. BIS Certification & Quality Control Order (QCO) Compliance Requirements

Regulatory Compliance Notice: Under Section 16 of the Bureau of Indian Standards Act, 2016, the Central Government issues Quality Control Orders (QCOs) making standard conformity and ISI marking mandatory for specific products. Manufacturing, importing, stocking, or selling non-certified goods listed under mandatory QCOs is a punishable offense.

5.1 Certification Schemes Applicable to Chemical Products

- Scheme I — Product Certification (ISI Mark):**
 - Requires factory inspection, verification of testing infrastructure, random sampling, and mandatory testing in BIS-recognized labs.
 - Applies to high-risk chemicals, packaged drinking water (IS 14543), safety glass (IS 2553), and critical industrial acids.
- Foreign Manufacturers Certification Scheme (FMCS):**
 - Enables overseas manufacturers producing chemicals for export to India to obtain an ISI mark license.
 - Requires appointment of an Authorized Indian Representative (AIR) and mandatory physical audit of the foreign factory.
- Scheme II — Compulsory Registration Scheme (CRS):**
 - Based on self-declaration of conformity supported by test reports from BIS-recognized laboratories. (Primarily used for IT/electronics, but evaluated for select specialty chemical applications).

5.2 Essential Key Requirements for License Grant

- **Scheme of Inspection and Testing (SIT):** The manufacturer must agree to strictly adhere to the BIS-defined SIT, maintaining complete batch-wise quality logs and calibration records.
- **In-House Testing Facilities:** Factory must possess calibrated testing equipment required for routine tests specified in the relevant IS.
- **Permissible Limits & Marking:** The BIS license number (CM/L-XXXXXX) and ISI logo must be prominently marked on packaging containers along with batch numbers and manufacturing dates.

6. Standard Status Tracking: Current, Revised & Superseded Standards

To prevent hallucination in AI response generation, RAG systems must track standard statuses accurately. Below are critical status mappings within CHD:

Standard ID	Standard Title	Current Status	Superseded / Remarks
IS 14543 : 2024	Packaged Drinking Water (Other Than Natural Mineral Water)	CURRENT	Replaced IS 14543:2004 (Revised to update safety parameters & limits).
IS 14543 : 2004	Packaged Drinking Water	SUPERSEDED	Withdrawn; replaced by 2024 edition.
IS 10500 : 2012	Drinking Water Specification	CURRENT	Reaffirmed regularly; incorporates Amendment 1 & 2. Replaced IS 10500:1991.
IS 2553 (Part 2) : 2019	Safety Glass - Road Transport	CURRENT	Replaced IS 2553 (Part 2): 1992. Mandatory under CMVR.
IS 252 : 2013	Caustic Soda, Pure and Technical	CURRENT	

Standard ID	Standard Title	Current Status	Superseded / Remarks
			Reaffirmed edition replacing 1991 version.

7. Product-to-Standard Mappings (RAG Mapping Table)

Product / Chemical Entity	Applicable Standard	Responsible Committee	Key Quality Parameters
Bottled Water / Packaged Water	IS 14543 : 2024	CHD 36	Microbiological purity, TDS, pesticide residue limits, heavy metals.
Caustic Soda (Lye / Flakes)	IS 252 : 2013	CHD 01	NaOH assay %, Sodium Carbonate content, Chloride, Iron content.
Sulphuric Acid	IS 266 : 1993	CHD 01	H ₂ SO ₄ concentration, residue on ignition, heavy metal limits.
Automobile Safety Windscreen Glass	IS 2553 (Part 2) : 2019	CHD 10	Optical distortion, light transmittance, impact resistance, fragmentation test.
Polyvinyl Chloride (PVC) Resin	IS 1047 : 2019	CHD 17 / CHD 01	K-value, bulk density, volatile content, particle size distribution.
Toilet Soap	IS 2888 : 2004	CHD 25	Total Fatty Matter (TFM) %, free caustic alkali, moisture content.
Industrial Oxygen Gas Cylinder	IS 309 : 2005	CHD 06	Oxygen purity % v/v, moisture content, carbon monoxide limits.

8. Frequently Asked Questions (FAQ) for AI System Training

Q1: What is the official standard for Packaged Drinking Water in India, and is BIS certification mandatory? Answer: The official standard for Packaged Drinking Water (other than natural mineral water) is IS 14543 : 2024. Yes, BIS certification is mandatory under Scheme I (ISI Mark) as enforced by Food Safety and Standards Authority of India (FSSAI) and Ministry Quality Control Orders. Manufacturing or selling packaged drinking water without a valid BIS license (CM/L number) is illegal.
Q2: Which BIS sectional committee handles standards for paints and protective varnishes? Answer: Standards for paints, varnishes, and related products are formulated and maintained by Sectional Committee CHD 20 (Paints, Varnishes and Related Products), while raw materials like pigments and resins are handled under CHD 21.
Q3: What is the difference between IS 10500 and IS 14543? Answer: IS 10500 : 2012 specifies requirements for <i>Drinking Water</i> supplied through public water systems / taps (guideline standard). IS 14543 : 2024 specifies standards for <i>Packaged Drinking Water</i> sealed in containers and intended for commercial sale, which requires mandatory BIS ISI certification and stricter hygienic packaging controls.

Q4: How can a foreign manufacturer obtain BIS certification for chemical products under CHD?

Answer: Foreign manufacturers must apply under the **Foreign Manufacturers Certification Scheme (FMCS)**. They must nominate an Authorized Indian Representative (AIR), submit an application along with manufacturing details, undergo factory inspection by BIS officers, and test product samples in authorized BIS laboratories in India.

9. RAG Retrieval Keywords, Synonyms & Taxonomy

To ensure accurate vector search retrieval, embedding models should map the following terms to the CHD department node:

Key Search Keywords: Bureau of Indian Standards, BIS CHD, Chemical Division Council, Chemical Department, Indian Standard, IS Number, Quality Control Order, QCO mandatory chemicals, Packaged Drinking Water IS 14543, Drinking Water IS 10500, Caustic Soda IS 252, Sulphuric Acid IS 266, Safety Glass IS 2553, Foreign Manufacturers Certification Scheme, FMCS ISI Mark, Scheme of Inspection and Testing, SIT BIS, Total Fatty Matter TFM, CHD 01, CHD 06, CHD 20, CHD 36, Heavy metal testing IS 3025.